NUIMPACT

IRM Impact Case Study Report

Fall 2025

Northeastern University

NUIMPACT
IMPACT INVESTING INITIATIVE

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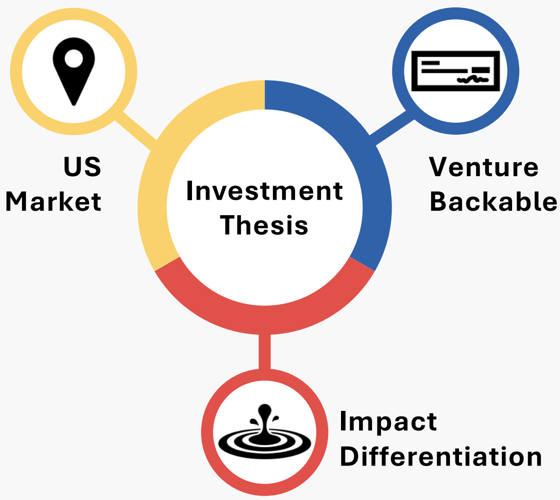
NUImpact Overview

- NUImpact is Northeastern University's student-led impact investing initiative and investment fund
- Since our inception in Spring 2016, the organization has served as a unique resource and thought exchange for the Northeastern community to understand purposeful capital, develop technical skills, and make targeted investments in companies that promote positive social and environmental impact

INSPIRE	EDUCATE	INVEST
Advance awareness of purposeful capital	**Connect** students to the ideas, practice, and careers in impact investing	**Invest** in mission-driven companies in the Northeast



Industry Focus

- **Education & Media**
- **Energy & Environment**
- **Finance & Technology**
- **Food & Agriculture**
- **Healthcare**

Our Semester-Long Investment Process

1	2	3	4	5
Prospecting and Sourcing	**Introductory Diligence**	**Deep Diligence**	**Term Sheet & Investment Pitch**	**Portfolio Management**
Analyst teams identify investment opportunities through pipeline **partner introductions**, online **research**, and attending community events.	Senior and Junior Analysts reach out to and conduct **calls with founders** to gauge interest and alignment with our **investment thesis**.	With feedback from **Fund Management Team & Advisors**, Senior Analysts lead drafting of **IRMS's (Investment Review Memorandums)** and pitch to Fund IC.	Analyst teams present IRM's to **Advisory Board** and then the **Investment Committee votes** to approve investment. Deal **terms** and documents passed to the Office of General Counsel for approval.	Director of Fund Operations and **Portfolio Management** communicate with founders, update models, and continue to **support** the company.

Education & Media: Robo Wunderkind

Problem Statement

Lack of early STEM & AI literacy in K–5 education

- Early exposure to coding, robotics, and AI is inconsistent—especially in under-resourced schools
- Limited access in Title I districts widens the digital skills gap
- Teachers often lack training, tools, and confidence to deliver hands-on STEM

70% of U.S. schools reached are classified as **Title I**

100+ districts in pipeline for new AI literacy curriculum

130+ schools and **40,000+** students currently using the program

Company Overview

Centralized STEM & AI Literacy Platform

- Modular robotics kits snap together in under 5 min
- Three coding levels for **ages 5-12**
- **100+ curriculum lessons** & updated literacy track
- Teacher PD, STEAM Hub collaboration tools, implementation support

Personalized Learning Benefits:

- **Students:** engaging robotics, early AI concepts, hands-on learning
- **Teachers:** low-barrier STEM implementation, confidence-building PD
- **Schools:** scalable curricula, high renewal rate due to classroom fit

Go-to-Market Strategy

Districts: "Land & Expand" strategy starting with pilots, expanding to additional grades/schools

Partnerships: With YMCA, Boys & Girls Clubs, ASAS

Homeschool: Licensing to gain access to homeschool markets

PD-Level: PD-led educator adoption model improves retention

More classrooms using robotics → higher student engagement → stronger district expansion

More districts adopting AI curriculum → stronger data & outcomes → improved product → faster renewals

More afterschool/homeschool pilots → broader brand presence → lower school acquisition costs

Impact Assessment

Impact Metrics

	Impact Metrics
Access & Inclusion	% of Title I schools served
Learning Time	Increase in student-centered STEM hours per week
Engagement	Student usage & project completion rates
Teacher Enablement	PD completion + self-reported confidence increases

Areas of Focus

Product Value
Hands on robotics + curriculum improves long-term STEM engagement

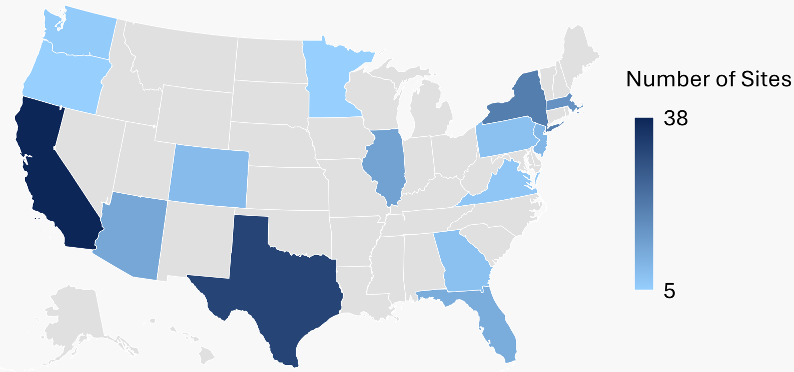
Inclusion and Access
70% of implementations are in Title 1; Low-barriers enable equitable participation

Inclusion in Education
Teachers without technical backgrounds can deliver high-quality STEM courses

Energy & Environment: THOLA

Problem Statement

Grid Stress Events / Year



Current State:
Large industrial buildings **lose $320K to $1.6M a year** to energy peak penalties

Inflated Energy Costs:
Price Energy Index **increased ~3%** over past 12 months

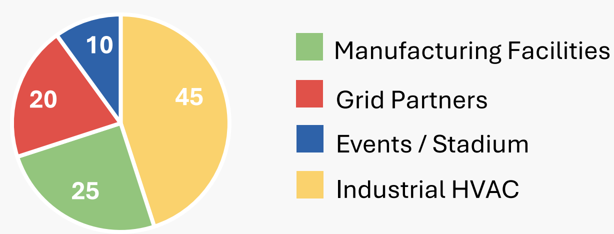
Effects on Underserved Communities:
Outages, high utility bills, grid instability forced upon surrounding communities

Company Overview

Solution

- THOLA's platform **monitors** energy consumption patterns, signals facilities to **reduce load** during peak demand
- Main **differentiator** being **cross-site coordination**: enabling regional balancing rather than isolated responses.

THOLA's Customer Segments (%)



Business Model

- THOLA's operates a **B2B subscription** platform for industrial and municipal clients
- The revenue model combines **software licensing** with **performance-based incentives** tied to energy savings and grid stabilization
- Partners with **utilities** and **city governments**, focusing on **scalability** during major events (i.e. 2028 LA Olympics, 2026 World Cup - Boston)

Go-to-Market Strategy

Industrial Sites → **Pilot Deployment** → **Partnerships** → **Regional Rollout**

Impact Assessment

	Impact Metrics	Target	Risk Factors
Reduced Grid Stress	Number of **Peak-events** avoided	**Prevent 5+ peak events** in first year	**Weather or grid issues** can lead to outages
Energy Cost Savings	**% decrease** in peak-time electricity costs	Reduce energy payments by **10–15%**	Cost savings **dependent** on **client response** and local utility
Underserved Communities	**Reduction** in **outage** hours in underserved areas	**10% fewer local outages** in first year	Reliant on grid **operator data** and **building cooperation**
Transparency in Data Sharing	% of users **providing anonymized** energy data	**80%+** take-up on shared energy	Certain companies may decline due to **privacy** or **competition**

Finance & Technology: Wallit

Problem Statement

A broken system for managing employee benefits

- The employee benefits ecosystem is **fragmented and complex**, leaving employees disengaged from programs that don't reflect their needs
- Multiple point solutions lead to fragmentation
- Low employee engagement
- Outdated platforms are a poor fit for modern & flexible workforces

61% of employees unsure about financial wellness benefits

80% Firm's belief in satisfaction

75% of employees stay longer with personalized benefits

58% Employee satisfaction

Company Overview

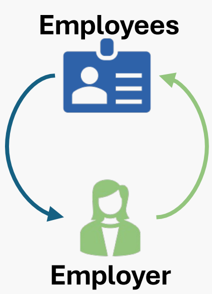
Solution

Centralized benefits: Connects employers and providers in one platform → Benefits Hub, Embedded Insurance/Payments

Simple reimbursement: Integrates financial tools for easy spending management

Personalized insights: Uses AI, spending data, and linked accounts to recommend benefits

Simplifies management and **boosts engagement** through programs



One app for **enrollment, reimbursements** and transparent **pricing**

Go-to-Market Strategy

Carriers → Wallit → Brokers → Employer → End Users

Access to Underserved Markets

Revenue + Data

Free Benefits

Unified Access

Network Effect Flywheel

- More wallets → better experience → increase loyalty
- More employers → economies of scale → stronger brand

Commercialization

- Supply: Enterprise Contracts with carriers
- Demand: Broker Licenses with national firms
- Broker Deployment

Selling Roadmap

- Train 200 Brokers (20 currently)
- Launch Payroll companies as second distribution channel and Scale team beyond founders

Impact Assessment

	Impact Metrics	Risk Factors
Product Value	**Renewal Ratio** (% of wallets or employer groups renewing each term)	Limited customer success bandwidth until team expansion; broker churn; onboarding friction for small employers
Inclusion and Access	**Wallet Composition** by Worker Type (% W2 vs. % 1099)	Brokers favor W2 employers; slower adoption among gig workers; limited direct-to-consumer reach
Inclusion in Education	**Educational Engagement** (availability and usage of benefits literacy content)	No current dedicated budget; partner dependency for content; low user engagement without incentives

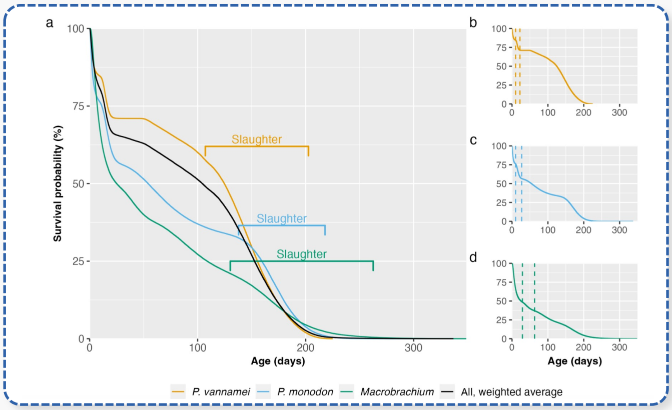
F&A: Knip

Problem Statements

Aquaculture farms face mortality events driven by disease, stress, and water-quality challenges

- Kaplan-Meier figure of major aquaculture species show **steep early mortality** (30-50% loss in the first 30-40 days)
- **Low harvest efficiency** (10-20% survival to slaughter age)
- Underscores systemic disease and stress challenges across global aquaculture
- In response, producers are shifting toward **natural**, **sustainable**, and **bio-based feed additives**
- Estimated CAGR of 4.6% (2023-2030) for US aquafeed market

Survival Curve for Major Aquaculture Species



Hannah McKay and William McAuliffe, 2024

Company Model

- Knip develops **Juv**, a naturally derived **microbial feed additive** that improves survival, growth, and stress resilience in farmed shrimp and fish
- Juv's proprietary leaf-based microbe produces bioactives that enhance gut and immune function while avoiding resistance risks seen with antibiotics or pesticides
- Demonstrated results: **+6-20pp higher survival**, **10-15% better feed conversion**, **28% higher farm margins**
- B2B feed mill partnerships: Juv added to existing formulas
- Direct sales to farms via demos and paid pilots
- Performance based recurring revenue per kg feed treated

Customer Verticals

Current

Farms
tilapia, trout, salmonid, and shrimp

Feed Mills

Future

Supplements
Human nutrition

Pet Food
Gut/immune support additives

Scalability

- Fermented by partners → low capex & COGS
- Designed to integrate easily into existing feed systems

Impact Assessment

	Impact Metrics	Risk Factors
Consumer Nutrition	Safer seafood through higher pathogen-free harvest rates	Limited transparency on farm antibiotic use restricts the ability to validate outcomes
Greener Aquaculture	Improved feed efficiency reduces resource use per kg of fish	Feed conversion gains may vary under inconsistent dosing or farm conditions
Product Value	Higher survival rates increase farmer profitability and harvest stability	Farmers may be slow to switch due to cost concerns or product unfamiliarity
Inclusion and access	Low-capex fermentation model improves affordability for small and mid-size farms	Smaller producers may still face financing limits that restrict adoption
Environmental Health	Cleaner fermentation lowers CO_{2e} intensity of production	Environmental benefits depend on regional energy mix and feedstock sourcing

Healthcare: LymeAlert

Problem Statement

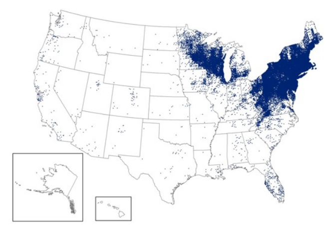
Lyme disease is prevalent in the United States, but effects can be mitigated through early detection

500k — Close to 500,000 U.S. **cases of Lyme disease every year**; highly concentrated in the northeast

35% — 35% of patients **receive treatment too late**, and current solutions are to wait for symptoms to appear

72hrs — Highest treatment success occurs when antibiotics are started within 72 hours of tick removal

Reported Cases of Lyme Disease in 2023



Lyme disease is the **most common vector-borne disease** in North America, leaving crippling financial and physical effects

Company Overview

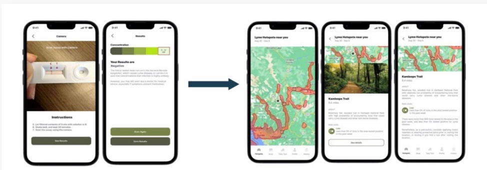
Solution

LymeAlert offers a simple, three-step process:

1. Users remove tick and send it for processing
2. Tick is tested and clear results are delivered
3. Users receive actionable treatment plan based on results

LymeAlert operates through a B2B2C Model

Mockup of LymeAlert App



Go-to-Market Strategy

Q3 2026 — Bulk, multi-tick test designed for high-volume stable and professional use to handle multiple ticks at once; enabling early diagnosis in horses

Q1 2027 — Affordable pack marketed for everyday pet protection, enabling early Lyme detection and rapid vet treatment to protect pets

Q1 2028 — Individual multi-disease tick test that supports quick decision-making after a bite, protects family health, and offers multi-pathogen analysis

Community Health Intelligence

LymeAlert employs training models for prediction and preventative treatment that ensure a focus on underrepresented regions in hot spot mapping

MilTICK	Bebop Labs
Military tick data sets	*NH Non-profit*

HIPPA

No HIPPA-sensitive data

Impact Metrics

Affordability	Lower average **cost per patient** (LymeAlert vs. standard clinic testing)
Patient Trust	**Net promoter score**
Accuracy	Higher % of patients **accurately diagnosed** early for Lyme disease
Health Improvement	% of patients who get **timely and appropriate treatment** after a positive LymeAlert test (within 7 days)

NUIMPACT
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360 Huntington Ave
Boston, MA 02115
United States

Email: leadership@nuimpactinvesting.com

Website: nuimpactinvesting.com

Social Media: @nuimpact